

Task ID	Phase	Module	Technical Task	Requirement / Why	Dependency	Priority	Definition of Done / Acceptance Criteria
P0-01	Phase 0	Architecture	Define solution architecture	Establish scalable architecture for marketplace, OMS, inventory, pricing, logistics, payments and AI so modules can evolve independently.	None	Critical	Architecture document, component diagram and API strategy approved
P0-02	Phase 0	Architecture	Define technology stack	Select frontend, backend, database, cache, object storage, messaging, monitoring and cloud based on scale, security and cost.	P0-01	Critical	Stack decision record approved
P0-03	Phase 0	Architecture	Define multi-tenant data model	Separate supplier, buyer, logistics and platform data while supporting one platform-wide transaction layer.	P0-01	Critical	ERD and tenant/data isolation rules approved
P0-04	Phase 0	Security	Identity and access design	Define OTP/password authentication, sessions, RBAC, admin privileges and audit trails.	P0-01	Critical	RBAC matrix and authentication design approved
P0-05	Phase 0	DevOps	Set up repositories and environments	Create dev, staging and production environments, CI/CD, branching and release process.	P0-02	Critical	Code deploys automatically to staging
P0-06	Phase 0	DevOps	Observability foundation	Centralise logs, metrics, error tracking, uptime monitoring and alerting before production launch.	P0-05	High	Alerts and dashboards working in staging
P0-07	Phase 0	Compliance	Compliance requirements matrix	Map applicable FSSAI, GST, e-invoicing/e-way bill, privacy, payments, food transport and state requirements to product workflows.	P0-01	Critical	Compliance checklist with owner and implementation status
P0-08	Phase 0	UX	Design system and UX flows	Create reusable UI components and user journeys for supplier, buyer, driver and admin apps.	P0-01	High	Clickable prototype and design system approved
P0-09	Phase 0	Integration	API standards	Define REST/GraphQL conventions, authentication, versioning, idempotency, error codes and webhook standards.	P0-01	High	API standards published
P0-10	Phase 0	Testing	QA strategy	Define unit, integration, API, security, performance, UAT and regression testing standards.	P0-01	High	QA strategy and test template approved
P1-01	Phase 1	Supplier	Supplier registration	Allow manufacturers/processors/approved suppliers to create accounts and submit business details.	P0-04,P0-08	Critical	Supplier can register and save onboarding application
P1-02	Phase 1	Supplier	Supplier KYC	Capture GST, PAN, FSSAI, bank, addresses, licences and authorised contacts; support document upload and verification status.	P1-01,P0-07	Critical	Supplier cannot go live until required KYC is approved
P1-03	Phase 1	Supplier	Supplier approval workflow	Admin can review, approve, reject, suspend and reactivate suppliers with reasons and audit trail.	P1-02	Critical	Every status change is logged
P1-04	Phase 1	Supplier	Supplier locations	Manage plants, warehouses, dispatch points and serviceable PIN codes.	P1-03	Critical	Inventory/order can be linked to a supplier location
P1-05	Phase 1	Catalogue	Product master	Create standardised chicken SKU structure: whole, cuts, boneless, frozen/chilled, pack sizes, weights, grades and attributes.	P0-03,P1-03	Critical	SKU can be created and reused across suppliers
P1-06	Phase 1	Catalogue	Supplier SKU listing	Supplier sets price, MOQ, lead time, availability, pack configuration and service areas for each approved SKU.	P1-05,P1-04	Critical	Approved SKU becomes purchasable only when mandatory fields are complete
P1-07	Phase 1	Catalogue	Batch and shelf-life fields	Capture batch/lot, production/processing date, expiry/best-before and storage conditions.	P1-05,P0-07	High	Batch data travels with inventory and order
P1-08	Phase 1	Inventory	Inventory ledger	Implement available, reserved, allocated, dispatched, delivered, rejected and expired inventory states.	P1-06	Critical	Inventory movements are immutable and auditable
P1-09	Phase 1	Inventory	Location-level inventory	Track stock by supplier location, SKU and batch.	P1-08,P1-04	Critical	Real-time stock shown per location
P1-10	Phase 1	Inventory	Inventory reservation	Reserve stock when an order is placed to prevent overselling.	P1-08,P1-19	Critical	Concurrent orders cannot oversell available inventory
P1-11	Phase 1	Pricing	Manual price management	Allow suppliers/admin to publish price, effective time, MOQ and buyer segment.	P1-06	Critical	Price changes are versioned with effective timestamps
P1-12	Phase 1	Pricing	Landed-price calculation	Calculate product + processing + packaging + logistics + taxes/fees where applicable to show buyer landed economics.	P1-11,P1-24	High	Buyer sees a transparent order total before checkout
P1-13	Phase 1	Buyer	Buyer registration	Register retailer, hotel, restaurant, QSR, caterer and institutional buyers.	P0-04,P0-08	Critical	Buyer account can be created
P1-14	Phase 1	Buyer	Buyer KYC	Capture GST/PAN/FSSAI/business details, locations, contacts and billing information.	P1-13,P0-07	Critical	Buyer verification status controls purchasing/credit
P1-15	Phase 1	Buyer	Buyer locations	Support multiple delivery addresses and operating hours.	P1-14	Critical	Order can select a verified delivery location
P1-16	Phase 1	Marketplace	Search and discovery	Search by SKU, brand, supplier, fresh/chilled/frozen, location and availability.	P1-05,P1-09,P1-13	Critical	Buyer can discover only active purchasable listings
P1-17	Phase 1	Marketplace	Product detail page	Show specification, pack size, MOQ, price, availability, delivery estimate, supplier and quality information.	P1-06,P1-11,P1-16	Critical	Buyer can make informed purchase decision
P1-18	Phase 1	Marketplace	Cart	Support multi-SKU cart and validate MOQ, stock, serviceability and price at checkout.	P1-16,P1-10	Critical	Invalid cart cannot proceed
P1-19	Phase 1	OMS	Order creation	Create order with buyer, supplier, location, SKU, quantity, price, taxes, delivery slot and payment terms.	P1-18,P1-12	Critical	Unique order ID and immutable order snapshot created
P1-20	Phase 1	OMS	Order state machine	Implement order states: pending, confirmed, processing, packed, dispatched, delivered, rejected, cancelled, refunded.	P1-19	Critical	Every transition is validated and audited
P1-21	Phase 1	OMS	Supplier order acceptance	Supplier can accept/reject/partially accept within SLA and system recalculates allocation.	P1-20,P1-10	Critical	Buyer receives status update
P1-22	Phase 1	OMS	Buyer order history	Show orders, invoices, status, delivery and reorder option.	P1-20	High	Buyer can repeat a previous order
P1-23	Phase 1	Notifications	Notification service	Email/SMS/WhatsApp/push abstraction for order confirmation, dispatch, delivery, rejection and payment events.	P0-09,P1-20	High	Templates and event logs available
P1-24	Phase 1	Payments	Payment gateway integration	Support online payment and payment status webhooks; use idempotency to avoid duplicate charges.	P0-09,P1-19	Critical	Payment status reconciles correctly after success/failure/retry
P1-25	Phase 1	Finance	GST invoice engine	Generate compliant invoice data/PDF with supplier, buyer, SKU, quantity, tax and order references.	P0-07,P1-20	Critical	Invoice generated against delivered/approved order
P1-26	Phase 1	Admin	Admin console	Manage users, suppliers, buyers, SKUs, prices, orders, disputes, notifications and configuration.	P1-03,P1-20	Critical	Admin can operate core marketplace without DB access
P1-27	Phase 1	Admin	Audit log	Record sensitive changes: KYC, price, inventory, order, payment, refunds and user permissions.	P0-04,P1-26	Critical	Admin can filter and export audit events
P1-28	Phase 1	Reporting	Core marketplace dashboard	GMV, orders, kg sold, active buyers/suppliers, cancellations, fulfilment and revenue.	P1-20,P1-24	High	Dashboard reconciles with transaction database
P1-29	Phase 1	Logistics	Manual delivery assignment	Admin can assign a delivery partner/driver and delivery slot to an order.	P1-20,P1-15	High	Assigned delivery is visible to all relevant parties
P1-30	Phase 1	Logistics	Proof of delivery	Capture OTP/signature/photo, delivery time, quantity accepted and rejection reason.	P1-29	Critical	Delivered order has immutable POD
P2-01	Phase 2	Procurement	RFO module	Buyer creates bulk RFO with SKU, quantity, location, delivery schedule, specification and validity.	P1-19,P1-05	Critical	RFO can be published to eligible suppliers
P2-02	Phase 2	Procurement	Supplier quotation	Suppliers respond with price, availability, MOQ, SLA and terms.	P2-01,P1-06	Critical	Quotes are comparable in standard format
P2-03	Phase 2	Procurement	Quote comparison	Compare supplier landed cost, fulfilment history, SLA and quality metrics.	P2-02,P1-28	High	Buyer/admin can compare quotes
P2-04	Phase 2	Procurement	Bulk demand aggregation	Aggregate similar buyer demand by city/SKU/date and create procurement batches.	P1-19,P2-01	Critical	Eligible demand is grouped without exposing confidential buyer data
P2-05	Phase 2	Procurement	Contract procurement	Create recurring supply contracts with volume, price formula, SLA and validity.	P2-02,P2-04	High	Contract generates scheduled POs
P2-06	Phase 2	OMS	Recurring orders	Automatically create orders from approved recurring procurement plans.	P2-05,P1-20	High	Failed generation triggers alerts
P2-07	Phase 2	Logistics	Logistics partner onboarding	Onboard 3PL, cold-chain fleets, local delivery partners and drivers with documents and service areas.	P1-03,P0-07	Critical	Only approved logistics partners can receive jobs
P2-08	Phase 2	Logistics	Route and job management	Create delivery jobs, assign vehicle/driver, pickup/drop sequence and SLA.	P2-07,P1-29	Critical	Job lifecycle mirrors order lifecycle
P2-09	Phase 2	Logistics	Driver app	Driver sees assigned jobs, navigation, pickup checklist, delivery checklist and POD capture.	P2-08,P1-30	Critical	Driver can complete delivery without admin intervention
P2-10	Phase 2	Logistics	GPS tracking	Track active vehicle location and ETA where enabled by partner/device.	P2-08	High	Admin sees latest location and stale-data status
P2-11	Phase 2	Cold Chain	Temperature data model	Store temperature readings against vehicle, trip, order, batch and timestamp.	P2-10,P0-07	Critical	Temperature history is retrievable per delivery
P2-12	Phase 2	Cold Chain	IoT integration interface	Create device/API adapter for temperature sensors and GPS providers.	P2-11,P0-09	High	External telemetry can be ingested and validated
P2-13	Phase 2	Quality	QC checklist	Create configurable checks for weight, packaging, temperature, batch, appearance and documentation.	P1-07,P2-11	Critical	QC result is attached to batch/order
P2-14	Phase 2	Quality	Digital batch traceability	Trace batch from supplier location through processing, dispatch, vehicle and buyer delivery.	P1-07,P2-13,P2-08	Critical	Single batch ID returns full trace
P2-15	Phase 2	Quality	Supplier performance scorecard	Calculate objective metrics: OTIF, rejection, shortages, complaints, temperature compliance and fulfilment.	P1-20,P1-30,P2-13	High	Metrics have defined formulas and date filters
P2-16	Phase 2	Disputes	Dispute centre	Buyer raises shortage/quality/delivery disputes with evidence and order references.	P1-30,P2-13	Critical	Dispute has owner, SLA and status
P2-17	Phase 2	Disputes	Credit note/refund workflow	Approve partial/full refund or replacement and create financial adjustment.	P2-16,P1-24,P1-25	Critical	Refund/credit note reconciles to original order
P2-18	Phase 2	Payments	Settlement engine	Calculate supplier payable after commission, logistics, adjustments and taxes/fees.	P1-24,P2-17	Critical	Settlement report reconciles to every order
P2-19	Phase 2	Finance	Reconciliation	Match gateway transactions, orders, invoices, refunds and supplier settlements.	P2-18,P1-25	Critical	Unmatched transactions appear in exception queue
P2-20	Phase 2	Marketplace	Supplier matching	Match buyer requirements to eligible suppliers using geography, SKU, stock, price and SLA.	P1-16,P1-09,P2-15	High	Matching results explain why suppliers qualify
P2-21	Phase 2	Pricing	City price board	Display current/previous price by city, SKU and product condition based on platform transactions and approved external data.	P1-11,P1-28	High	Price history is timestamped and source-labelled
P2-22	Phase 2	Analytics	Buyer procurement dashboard	Show spend, kg, average price, supplier mix, fulfilment, quality and savings.	P1-28,P2-15	High	Buyer can filter by date/location/SKU
P2-23	Phase 2	Analytics	Supplier dashboard	Show orders, demand, stock, fulfilment, revenue, price and quality metrics.	P1-28,P2-15	High	Supplier sees only its authorised data

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P2-24	Phase 2	Commerce	WhatsApp ordering	Allow approved buyers to search, quote, confirm and track orders through WhatsApp using approved provider APIs.	P1-16,P1-19,P1-23	High	WhatsApp order maps to same OMS order ID
P2-25	Phase 2	Commerce	Reorder engine	One-click or WhatsApp reorder using prior SKU, quantity and delivery preference with current price validation.	P1-22,P2-24	Medium	Old price is never reused without validation
P3-01	Phase 3	AI	Demand forecasting	Forecast SKU/city/day demand using transaction history, seasonality, day-of-week and events.	P1-28,P2-04	Critical	Forecast accuracy tracked against actuals
P3-02	Phase 3	AI	Supply forecasting	Forecast expected supplier availability and identify future shortages/surplus.	P3-01,P1-09	High	Supplier-level forecast shown with confidence
P3-03	Phase 3	AI	Dynamic pricing recommendation	Recommend price ranges using supply, demand, inventory age, geography, logistics and historical transactions; human approval initially.	P3-01,P2-21,P2-18	Critical	Every recommendation has inputs, timestamp and approval status
P3-04	Phase 3	AI	Procurement optimisation	Recommend supplier allocation for aggregated demand considering price, SLA, capacity and risk.	P2-20,P3-01,P2-15	High	Recommendation is explainable and can be overridden
P3-05	Phase 3	AI	AI procurement assistant	Natural-language assistant for buyer queries, sourcing and order creation with confirmation before committing.	P2-24,P3-03,P3-04	High	Assistant cannot place order without authenticated buyer confirmation
P3-06	Phase 3	AI	Exception detection	Detect unusual shortages, price spikes, late deliveries, rejection patterns and temperature breaches.	P2-15,P2-11,P2-19	High	Exceptions generate actionable alerts
P3-07	Phase 3	Credit	Buyer credit profile	Create transaction-based buyer risk profile using platform order, payment and dispute history; avoid unsupported inferences.	P2-19,P2-22	Critical	Profile shows only permitted/verified data and explainable metrics
P3-08	Phase 3	Credit	Credit partner integration	Integrate approved financing/credit providers through APIs and consent-based data flows.	P3-07,P0-07	Critical	Credit decision and status are securely exchanged
P3-09	Phase 3	Credit	Credit limit engine	Calculate platform purchase limit using configured partner decisioning and buyer history.	P3-07,P3-08	High	Limits and expiry are auditable
P3-10	Phase 3	Collections	Payment reminder automation	Automate invoice reminders via WhatsApp/SMS/email/voice based on due dates and buyer preferences.	P2-19,P1-23	High	Every communication is logged with opt-out controls
P3-11	Phase 3	AI	Voice ordering	Convert authenticated voice requests into draft orders with language support and confirmation.	P2-24,P3-05	Medium	Voice order requires confirmation before placement
P3-12	Phase 3	AI	AI quality assistant	Summarise QC/dispute evidence and recommend workflow action; human approval required for financial decisions.	P2-16,P2-13	Medium	Evidence links are retained
P3-13	Phase 3	Analytics	National control tower	Create HQ dashboard for GMV, kg, supply, demand, price, fulfilment, logistics, quality, credit and exceptions.	P3-01,P3-06,P3-09	Critical	City and national views reconcile
P3-14	Phase 3	Data	Data warehouse	Create analytics warehouse/lakehouse for historical transactions and BI without impacting OLTP performance.	P0-02,P1-28	High	ETL/ELT freshness SLA defined
P3-15	Phase 3	Data	Data governance	Define ownership, retention, consent, PII classification, access, deletion and audit rules.	P0-07,P3-14	Critical	Data catalogue and access matrix approved
P3-16	Phase 3	Integration	ERP/accounting APIs	Integrate buyer/supplier ERP/accounting systems for orders, invoices, inventory and reconciliation.	P0-09,P1-25,P2-19	High	Sync errors are visible and retryable
P4-01	Phase 4	Marketplace	Digital auction engine	Enable controlled supplier bidding for qualified RFQs with rules, bid windows and audit trail.	P2-01,P2-02,P3-03	High	Auction closes deterministically and records bid history
P4-02	Phase 4	Marketplace	Forward procurement	Support future-date volume commitments and contract pricing subject to legal/commercial rules.	P2-05,P3-01	Medium	Commitments have capacity and cancellation rules
P4-03	Phase 4	Network	City micro-hub management	Manage cross-docking/micro-hub inventory, capacity, inbound/outbound and service radius.	P2-08,P2-14	High	Hub inventory reconciles with supplier and delivery records
P4-04	Phase 4	Network	Multi-city optimisation	Optimise supply allocation across cities using forecast, capacity, transport cost and SLA.	P3-02,P3-04,P4-03	High	Recommendations show cost/service trade-off
P4-05	Phase 4	Finance	Escrow/payment orchestration	Implement regulated payment/escrow structure only with appropriate licensed partners and legal review.	P2-18,P0-07	Critical	Money movement design approved by legal/compliance
P4-06	Phase 4	Finance	Insurance integration	Offer shipment/product/credit insurance through licensed partners where commercially viable.	P2-08,P3-09	Medium	Policy status linked to relevant transaction
P4-07	Phase 4	Data	Poultry price index	Create transparent methodology for an aggregated platform price index by product and city.	P2-21,P3-14	Medium	Index methodology and source coverage documented
P4-08	Phase 4	API	Buyer procurement API	Expose secure APIs for large hotels/QSRs to send forecasts, create POs and receive status/invoice updates.	P3-16,P0-09	High	API authentication, rate limits and audit logs implemented
P4-09	Phase 4	API	Supplier supply API	Allow large manufacturers to push inventory, price, batch and dispatch updates directly from ERP.	P3-16,P1-09	High	Near-real-time sync with retry/dead-letter handling
P4-10	Phase 4	Platform	Partner marketplace	Onboard approved logistics, finance, insurance, cold-chain and technology partners with standard APIs.	P2-07,P3-08,P4-06	Medium	Partner lifecycle and permissions managed centrally
P4-11	Phase 4	Platform	Scalability and performance hardening	Load test critical APIs and optimise for peak ordering periods, city expansion and high transaction volume.	P3-14,P4-08,P4-09	Critical	Performance targets met under defined load
P4-12	Phase 4	Security	Security testing and certification readiness	Conduct VAPT, dependency scanning, secrets management, encryption review, access review and incident response testing.	P0-05,P3-15,P4-11	Critical	Critical vulnerabilities resolved before production expansion

Phase	Purpose	Key Deliverables	Exit Gate
Phase 0 – Foundation	Architecture, security, compliance, DevOps and UX foundation	Architecture, ERD, APIs, environments, RBAC, compliance matrix, QA strategy	Approved architecture + staging environment
Phase 1 – MVP	Direct B2B marketplace transaction flow	Supplier/buyer onboarding, catalogue, inventory, pricing, orders, payments, invoices, basic logistics, POD	Pilot can transact end-to-end in one city
Phase 2 – Operations	Make the platform operationally powerful	RFQ, demand aggregation, contracts, logistics app, GPS, cold chain, QC, traceability, disputes, settlement, WhatsApp	Pilot reaches repeat B2B usage with measurable fulfilment
Phase 3 – Intelligence	Add AI, credit and control-tower capabilities	Forecasting, price recommendations, procurement AI, credit, collections, voice, warehouse, national dashboard	AI recommendations and finance workflows operate with controls
Phase 4 – Network Scale	Create national network effects and infrastructure	Auction, forward procurement, micro-hubs, multi-city optimisation, APIs, partners, index, scale/security	Platform ready for multi-city/pan-India expansion

Dependency Area	Must Exist Before	Why
Identity & RBAC	Supplier/buyer/admin modules	Every transaction must be attributable to an authorised actor.
Master data / SKU	Catalogue, inventory, pricing, orders	Orders and inventory require standard product definitions.
Inventory ledger	Checkout/order reservation	Prevents overselling and provides auditable stock movement.
OMS	Payments, logistics, notifications, reporting	Order is the central transaction object.
Payments	Settlement, refunds, credit	Financial modules need a canonical transaction reference.
Logistics	GPS, cold chain, POD	Tracking data must attach to a delivery job/order.
Batch/QC	Traceability, disputes	Quality evidence needs batch and order linkage.
Data warehouse	Advanced AI and national BI	Historical, clean data is required for forecasting and analytics.
Consent/compliance	Credit, messaging, external integrations	Sensitive data and regulated workflows require controls first.

Priority	Include in MVP	Reason	Do Not Build Initially
Must Have	Supplier + Buyer KYC	Creates trusted B2B network	Complex public marketplace seller onboarding
Must Have	SKU catalogue + inventory	Core supply visibility	AI price optimisation
Must Have	Manual/dynamic supplier pricing	Allows transactions without complex pricing AI	Auction/forward contracts
Must Have	Order Management	Core transaction engine	Autonomous AI ordering
Must Have	Payments + invoices	Commercial closure	Full escrow orchestration
Must Have	Basic delivery assignment + POD	End-to-end fulfilment	Full multi-city route optimisation
Must Have	Admin/control dashboard	Operate pilot without engineering support	National control tower
High	WhatsApp ordering	India B2B adoption friction reduction	Voice ordering
High	Basic RFQ	Validate institutional procurement	Automated procurement optimisation
Later	IoT cold-chain	Important at scale but depends on logistics partners/devices	Custom hardware
Later	Credit engine	Requires transaction history and regulated partners	In-house lending
Later	AI forecasting/pricing	Needs clean historical data	Fully autonomous price setting